

Subject: Program objectives review for the eight ABET accredited programs at the CoE

Key Notes:

1. On Oct 23rd: We have received the dean approval to keep the CoE mission as is. So, ABET would expect your program objectives to be mapped to the following (key words from our mission statements):

- Address the needs of the State and nation
- Educate students for careers
- Leadership
- Innovation
- Expand the base of engineering knowledge
- Original research
- Developing technology
- Serve Society needs
- Service to industry, government and the engineering profession

2. We have reviewed the **program objectives** for our eight ABET accredited programs based on 7 criteria. Points of improvements have been summarized in a table and could be shared with the committee.

3. A brief report that includes the analysis, recommendations and actions related to program objectives for all eight programs will be sent to the department dedicated assessment chair today.

4. Office of Programs and Academic Assessment at the university has adopted an assessment process that all programs, in the east and west, campus must follow. The university has adopted an online assessment tool (TK20) to report the assessment process in each program. College of Engineering will have a training on Wednesday Nov 28th at 1:30 in SEO 1000 to a) learn the new process of assessment at the university level, b) learn how to use TK20 to report your program assessment components. This is a one hour session that will be run by the university director of academic assessment, Prof. Ranfen Li and CoE Educational Assessment Director, Prof. Loay Abusalah. All DUGS as well selected office assistants are required to attend.

5. The initial results from a short survey (one to one interview) conducted by our College Education Assessment office shows that more than 45% of our faculty can't list one program objective, more than 70% can't list two program objectives while most of the engineering faculty did not read the college mission in the last three years. After finalizing the program objectives, we recommend to have the program objective, posted in a large poster, at each dedicated department floor. This will help faculty to better know their program objectives. We can do the same for the college mission.

6. Dr. Abusalah, will take over the process of collecting the assessment results for the program objectives starting from next semester. The analysis, recommendations and actions are still needed to be done through the DUGS (or the dedicated department assessment chair). Please keep doing what you need to do tell your DUGS received a confirmation from Loay. This might take few months as Loay has to understand the assessment process in each department one by one before taking over.

CoE Mission:

"The Mission of the UIC College of Engineering is to address the needs of the State and nation through excellence in education, research and service; to educate students for careers of leadership and innovation in engineering and related fields; to expand the base of engineering knowledge through original research, developing technology to serve the needs of society; and to benefit the public through service to industry, government and the engineering profession."

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Program Objectives analyses for the eight programs in the college of engineering at UIC

The following criteria have been used to evaluate the objectives for the eight ABET programs at the college of engineering:

1. The mapping of the program objectives to the CoE mission.
2. The wording of the objectives statements is following ABET standards (Bloom's taxonomy).

Program objectives are broad statements that describe the career and professional accomplishment that the program is preparing graduates to achieve. Some of the verbs we can use are: APPLY A RULE, ASSESS, CLASSIFY, COMPOSE, CONSTRUCT, DEFINE, DEMONSTRATE, DESCRIBE, DIAGRAM, DISTINGUISH, ESTIMATE, EVALUATE, IDENTIFY, INTERPRET, LABEL, LOCATE, MEASURE, NAME, ORDER, PREDICT, REPRODUCE, SOLVE, STATE A RULE, TRANSLATE.

3. Assessing the program objectives using the university five assessment process: measure, results, analysis, recommendations, and actions.

Office of Programs and Academic Assessment at the university has adopted an assessment process that all programs, in the east and west, campus must follow. The university has adopted an online assessment tool (TK20) to report the assessment process in each program. College of Engineering will have a training on ??? in SEO 1000 to a) learn the new process of assessment at the university level, b) learn how to use TK20 to report your program assessment components. This is a one hour session that will be run by the university director of academic assessment, Prof. Ranfen Li. All DUGS as well selected office assistants are required to attend.

4. The availability of the program objectives in the web.

Program objectives, student outcomes, and the course that are used to assess the student outcomes are required to be published online.

5. Department mission statement

The department mission statement is not required by ABET. However, if you decide to have one, then it has to be consistent with the CoE mission.

6. The availability of student outcomes information to the students.

7. The mapping of the program objectives to student outcomes.

	1. CoE mission to Program objectives	2. Program Objectives Wording	3. Assessment Process	4. Web Ease	5. Dept Mission	6. Student outcomes	7. Program objectives to student outcomes map
Bio Eng	NI	G	NA	G	NI	NI	NI
Chemical Eng	NI	G	NA	NI	G	G	G
Civil Eng	NI	G	NA	NI	NA	G	NI
Computer Science	NI	NI	NA	NI	NA	NI	NI
Computer Eng	NI	G	NA	NI	NA	NI	NI
Electrical Eng	NI	G	NA	NI	NA	NI	NI
Mechanical Eng	NI	NI	NA	NI	NA	G	NI

Industrial Eng	NI	NI	NA	NI	NA	G	NI
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Legend:

NI- Need Improvements.

G- Good as is.

NA: Not Announced.

Chemical Engineering Department

The mission of the Department of Chemical Engineering is to serve society through excellence in education, research, and public service. We provide for our students an education in chemical engineering, and we aspire to instill in them the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers. Through scholarship, the Department strives to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond.

Program Educational Objectives

1. Depth: In engineering practice or advanced study, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, and creative design.
2. Breadth: Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3. Professionalism: Graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics and professionalism.
4. Life-long learning: Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will emerge with an appreciation for and effectiveness in life-long learning.

Program Outcomes with ABET criteria (a) - (k) and Program Educational Objectives

#	Description	ABET	PEO
1	Apply knowledge of mathematics and science in engineering problems	a	1
2	Identify and formulate engineering problems	e	1
3	(3) Use techniques, skills, and modern chemical engineering tools	k	1
4	Design experiments	b	1
5	Conduct experiments	b	1
6	Analyze and interpret data	b	1
7	Design engineering systems toward technical specifications	c	1
8	Refine engineering designs within realistic constraints (economic, environmental, social, ethical, health and safety, manufacturability and sustainability)	c	2
9	Understand the societal and broader impact of	h	2

	engineering solutions		
10	Understand the significance of contemporary issues	j	2
11	Recognition of and engagement in life-long learning	i	4
12	Function effectively in multidisciplinary design teams	d	3
13	Communicate engineering reasoning and solutions in a professional and effective manner	g	3
14	Understand ethical and professional responsibilities	f	3

Analysis

- The department mission is consistent with the college mission.
- The term "Educational objectives" is not used anymore.
- Program objectives are not mapped to the college mission very well. Leadership, serving the society, original research, as well as service to industry, government and the engineering profession have not been directly addressed in the objectives.
- The wording used to write the program objectives are well chosen and is following the Blooms taxonomy.
- The assessment link online is really rich with great assessment information. however, it is NOT easy for a student to find the program objectives in your department website.
- The student outcomes are listed in the department website. However, it is not written as a statement.
- Courses to student outcomes map is available in the department website. Good Job! We will review the content of the mapping later this year.

Recommendations (Due: Jan 15th 2013)

- Program Educational objectives are now "Program Objectives".
- Rewrite your program objectives to directly address the leadership, serving the society, original research, and services to industry, government and the engineering profession.
- List your program objectives under "About us> our department".
- List the student outcomes of your program under About us> our department. You can keep the tables under "Assessment".

Actions (Due: Dec 14th, 2012)

1. What are the methods used by your program to assess the program objectives? (for example: exit interview, alumni survey, expo survey, ...).
2. Please include a copy from all the surveys used to assess the program objectives.
3. When do you conduct each survey?
4. Who is responsible to conduct each survey (or any other assessment component you use to assess your program objectives).
5. Do you have an assessment committee for your program? If yes, please list faculty names.

6. Do you (or assessment committee) **analyze** and record the collected results to find point of improvements? If yes, please list any point of improvement you have noticed by reading the survey results (or any other assessment component you are using).
7. Do you take **recommendations** and **actions** to improve the outcomes of your program objectives? If yes, please list an example for a recommendation and an action taken by your department to improve the outcomes of your program objectives.
8. Please send me a copy from your 2008/ 2009 program Self Study Report (SSR)

Survey Statistics (ongoing...)

- 31 CoE faculty
- 4 lecturers, 12 Assistant, 5 associate, 7 Full
- All full timers

Survey Questions

1. List one Program Objective
2. List two program objectives

3. List three program objectives

4. Are you aware of the method used by department to measure program objectives?

5. Did you see/read the CoE mission anytime in the last two years?

6. Would you like to change it?

Question	1	2	3
Lecturers	3/4	1/4	1/4
Assistant	4/12	2/12	0/12
Associate	4/5	3/5	3/5
Full	6/7	3/7	2/7
Question	4	5 (YES)	6 (YES)
Lecturers	4/4	0/4	NA
Assistant	5/12	3/12	NA
Associate	3/5	4/5	3/5
Full	7/7	4/7	4/7